Phytochemical, Toxicological and Haematological Potentials of *Dacryodes edulis* Seed in Carbon Tetrachloride Exposed Rats

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ABSTRACT

This study evaluates the phytochemical and toxicological potentials of ethanolic seed extract of *Dacryodes edulis* (*DE*) and its effect and hematological profile of Carbon tetrachloride (CCl_4) exposed male rats. Phytochemical studies were carried out on the seed and acute toxicity was determined using Lorke method. Wistar rats (N=90; 180-200g) were distributed into 3 groups; control, CCl_4 exposed (IP 1ml/kg in 1:1 olive oil every 72 hours in 3 doses) and CCl_4 exposed treated *DE* (1500 mg/kg). Blood samples were collected at days 1, 7, 14, 21 and 28 for the hematological variable herein investigated. The data were analyzed using ANOVA (SPSS Version 17.0) and LSD post hoc test at $p \leq 0.05$. Qualitative and quantitative phytochemical composition showed *DE* seeds to be a rich source of carbohydrate, Phenol, Alkaloid, Flavonoid and Tannins and safe at dose of 5000mg/kg body weight. CCl_4 induced significant increase in white blood cell and neutrophil counts but decreased in red blood cell, hematocrite, platelet and lymphocytes counts. Treatment with *DE* resulted in mitigating the observed CCl_4 induced alterations. Hence, *DE* possess phytochemical component that may improved CCl_4 induced haematological alterations in rats.

Keywords: *Dacryodes edulis*, Phytochemicals, Toxicological, Haematological, Rats.

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1. INTRODUCTION

Exposure to environmental pollutants, toxic chemicals, and drugs have been reported to cause cellular injuries [1]. Carbon tetrachloride (CCl_4) is one of the xenobiotics that has been reported to induce acute and chronic tissue injuries or necrosis [2, 3], through lipoperoxide and free peroxide radical generation [4, 5]. CCl_4 has been used extensively in animal models to cause kidney, heart, testis, liver and brain pathogenesis [6-9] and inflammatory response [10].

Plants are important sources of medicines and presently about 25% of pharmaceutical prescriptions in the United States contain at least one plant-derived ingredient [11]. *Dacryodes edulis* (*DE*) contains a wide range of bioactive compounds such as alkaloids, terpenes, flavonoids, tannins, thiamine and saponins [12]. The plant has long been used in the traditional medicine for the treatment of various ailments such as wound, skin diseases, dysentery and fever [12]. The seeds of *DE* are used traditionally as a remedy for gastrointestinal disorders like diarrhoea, dysentery [13]. The extract has been found to show biological activities such as anti microbial, anti oxidant and anti sickle cell anaemia [14]. Hence, the aim of this study is to investigates the proximate and phytochemical components, acute toxicity and haematological potential of ethanolic extract of *DE* following CCl_4 exposure in adult male Wistar rats.

2. MATERIALS AND METHODS

2.1. Plant material and preparation: The *DE* seeds were collected around Awka, Anambra State, Nigeria during May - June and authenticated by a Botanist, in the Department

of Botany, Ambrose Alli University, Ekpoma, Edo State, Nigeria. The seed were shade-dried and thereafter pulverized. 850g of the pulverized seeds was soaked in 70% ethanol for 4 days. The ethanol mixture was filtered to obtain the ethanolic extract according to the method of Akinpelu *et al.,* [15]. The residue was re-extracted, the same solvent in 24 hours. The mixture was concentrated in-vacuo using a rotary evaporator to remove the ethanol [16]. The resultant was then used for the experimental work.

2.2. Proximate analysis of the plant seed: The proximate analysis to determine the moisture content, crude protein, fibre, crude fat, ash and carbohydrate in the *DE* seeds was conducted using AOAC [17] method. The analysis was in triplicates.

2.3. Phytochemical analyses: Phytochemical studies were carried out following standard laboratory proceducres.

2.4. Oral acute toxicity study: Acute toxicity (LD_{50}) study was carried out by the method described by Lorke [18] with minor modifications in the dosage of extract administered.

2.5. Study design: Adult male Wistar albino rats (weighing 180-200g) were sourced from the animal house, College of Medicine, Ambrose Alli University, Ekpoma and housed in clean cages and fed with normal pellet and water *ad libitum*. They were acclimatized to the laboratory conditions for 2 weeks. The rats were randomly divided into 3 groups as normal control, negative control treated with 1ml/kg body of CCl_4 in olive oil (at a ratio of 1:1; intraperitoneally injection) every 72 hours for 3 doses and test group receiving 1500 mg/kg *DE* extract (via intragastric route) after CCl_4 intoxication. Six rats were sacrificed from each group on days 1, 7, 14, 21 and 28 post treatment.

2.6. Estimation of blood biochemical parameters: At the end of the *DE* extract treatment period, the animals were anaesthetized with chloroform and blood samples collected through cardiac puncture for hematological analysis.

2.7. Hematological parameters: Direct Current **(**DC) sheath flow detection method by Sysmex analysers was used to count red blood cells and platelets. 0.5ml of blood sample was measured into the sample chamber of the machine using a syringe. The values were then read from the graphical displays of distribution curves generated.

2.8. Statistical analysis: Data were analyzed using SPSS (Version 17.0) software and the results were expressed as Mean ± SEM after performing one-way analysis of variance (ANOVA) followed by Least Significant Difference (LSD) test for comparison between two groups. The p-values less than 0.05 were considered statistically significant.

3. RESULTS

1. Proximate and phytochemical analysis of the seed of *Dacryodes edulis*

Figure 1 shows shows the percentage proximate composition of *DE* seed and was observed to contains soluble carbohydrate (42.16±0.09%), moisture (23.00±0.04%), nitrogen (20.19±0.48%), lipid (7.51±0.18%), crude fibre (4.76±0.10%) and ash (2.38±0.09%).

Table 1 shows the result of the qualitative and quantitative phytochemical composition of *DE* seeds. Qualitative phytochemicals include flavonoid, alkaloid, steroid, tannins, Saponin, phenol, phytosteroids, cardiac glycosides, terpenoid and anthraquinones. The seed was high in phenols (261.82±0.54 mg/100g), alkaloid (96.00±0.02 mg/100g), tannins (42.98±0.09 mg/100g), flavonoid (38.00±0.04 mg/100g) and cardiac glycosides (21.00±0.31 mg/100g) with low amounts of saponin (1.65±0.06mg/100g).

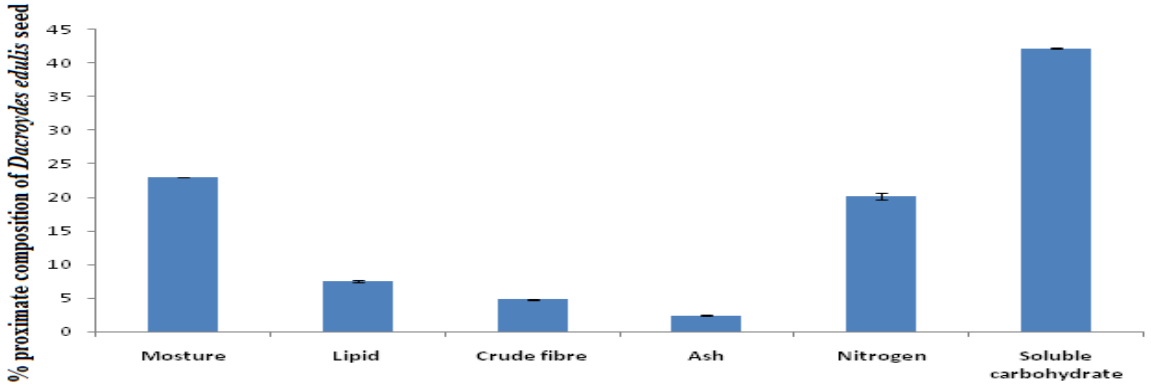


Figure 1. The percentage proximate composition of *Dacroydes edulis* seeds.
Values are mean ± SEM of triplicate determinations

Table 1. The qualitative phytochemical composition of *Dacryodes edulis* seeds

Phytochemical composition	Qualitative characteristic	Quantitative values (mg/100g)
Anthraquinones	+	0.65±0.06
Phlobalanin	-	ND
Terpenoid	+	1.70±0.11
Anthrocyanin	-	ND
Alkaloid	+	96.00±0.02
Flavonoid	+	38.00±0.04
Tannins	+	42.98±0.00
Saponins	+	1.65±0.06
Phenol	+	261.82±0.54
Phytosteroids	+	10.80±0.18
Cardiac glycosides	+	21.00±0.31

Where "+" indicates "present" and "–" indicates "not present, Values are mean ± SEM of triplicate determinations ND= not determined.

2. LD_{50} (lethal dose) of ethanolic seed extract of *Dacryodes edulis*

Table 2 presneted the oral acute toxicity study of ethanolic seed extract of *DE*. The result showed that the *DE* extract at the treated doses of 10mg/kg to 5000mg/kg was not toxic to the rats considering that no death or alteration in physical or behavior occurred after 24 hrs of continuous monitoring.

3. Red blood cells and platelet indices following ethanolic seed extract of *DE* treatment in CCl_4 exposed rats

Figure 2 shows the effect of ethanolic seed extract of *DE* administration on red blood cells and platelet indices in CCl_4 exposed rats. Exposure to CCl_4 resulted in significant decrease in RBC, HCt, platelet and MPV while treatments with *DE* resulted in significant duration dependent increase in these parameters compared with the CCl_4 exposed group.

4. White blood cell and differentials following ethanolic seed extract of *DE* treatment in CCl_4 exposed rats

Figure 3 shows the effect of ethanolic seed extract of *DE* administration on white blood cells and differentials in CCl_4 exposed rats. Exposure to CCl_4 resulted in significant

increase in WBC and neutrophil but decrease in lymphocytes while treatments with *DE* resulted in significant duration dependent changes in these parameters compared with the CCl_4 exposed group

Oral acute toxicity study of partially purified ethanolic seed extract of Dacryodes edulis

Groups	N	Dose (mg/kg body weight)	Mortality	Percentage mortality	Other symptoms
Phase 1					
A1	3	10	Nil	0.00	Nil
B1	3	100	Nil	0.00	Nil
C1	3	1000	Nil	0.00	Nil
Phase 2					
A2	1	1500	Nil	0.00	Nil
B2	1	3000	Nil	0.00	Nil
C2	1	5000	Nil	0.00	Nil

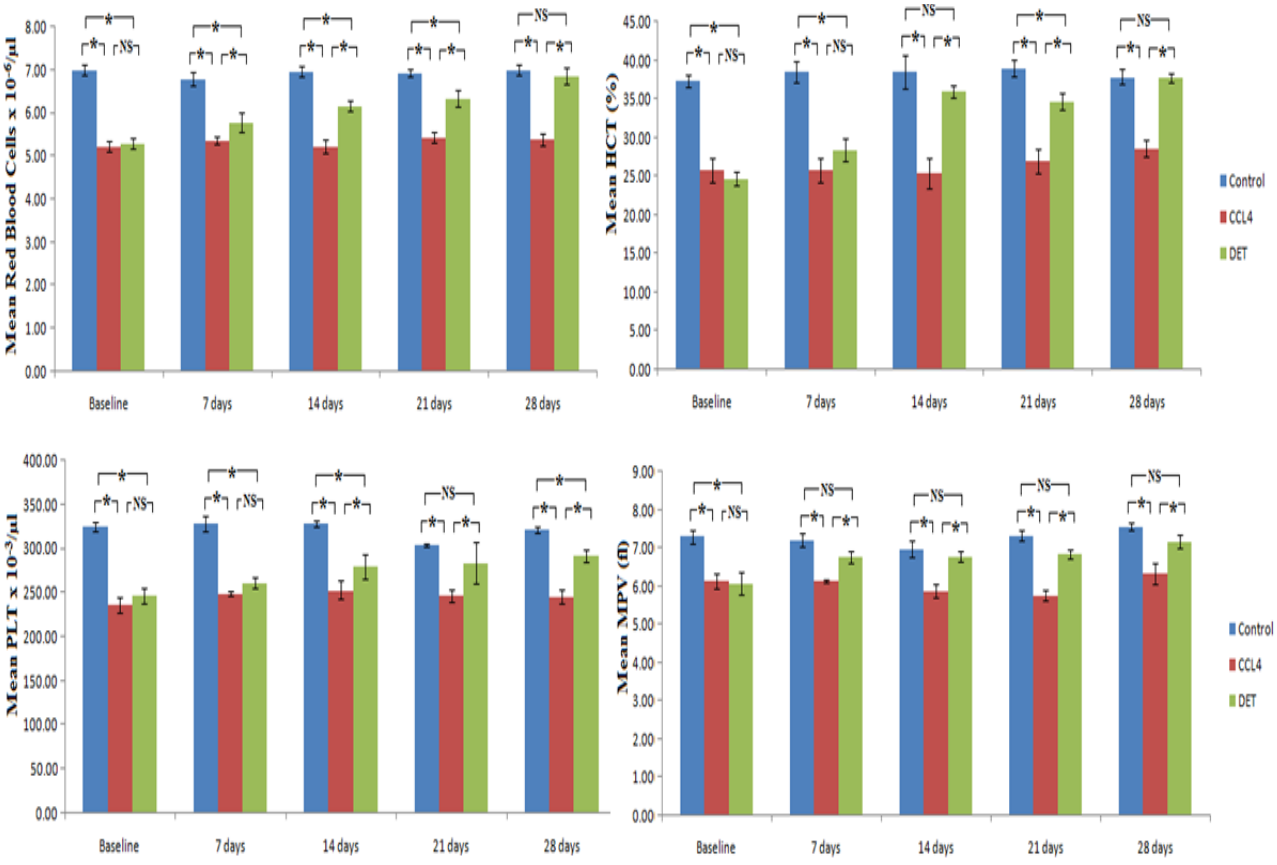


Figure 2. Effect of ethanolic seed extract of *Dacryodes edulis* on (a) red blood cells, (b) HCT, (c) platelet and (d) MPV in carbon tetrachloride exposed rats

Values are mean ± SEM, *significant difference at $p<0.05$, NS = not significant, CCl_4 = untreated Carbon tetrachloride exposed groups, DET = CCl_4 exposed treated *Dacryodes edulis*, Control= Unexposed untreated groups.

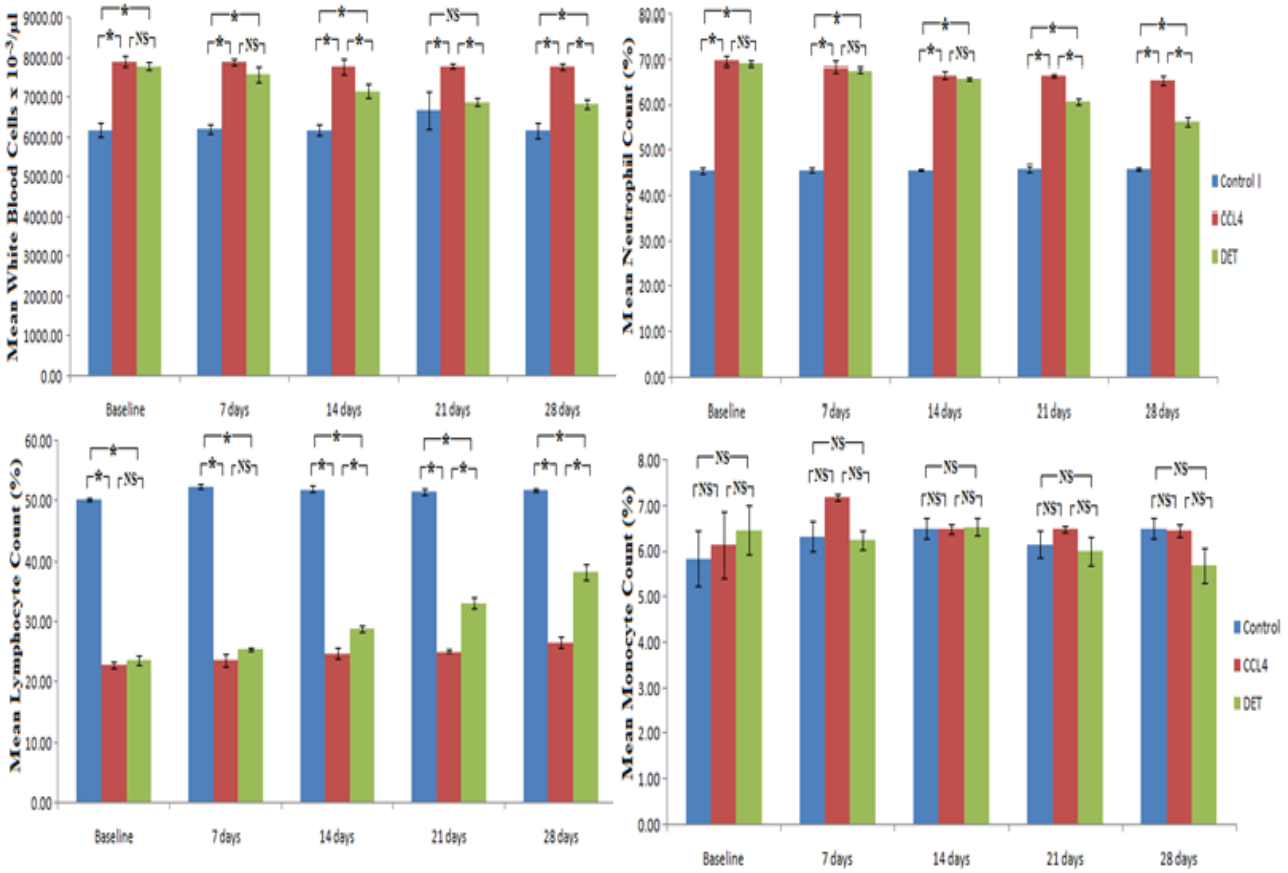


Figure 3. Effect of ethanolic seed extract of *Dacryodes edulis* on (a) white blood cells, (b) neutrophil, (c) lymphocyte and (d) monocytes in carbon tetrachloride exposed rats

Values are mean ± SEM, *significant difference at $p<0.05$, NS = not significant, CCl_4 = untreated Carbon tetrachloride exposed groups, DET = CCl_4 exposed treated *Dacryodes edulis*, Control= Unexposed untreated groups.

4. DISCUSSION

In the present study, effort was made to ascertain the nutritive composition and acute oral toxicity of the seed of *DE* and its effect on some haematological parameters in CCl_4 exposed rat model. The result on proximate, qualitative and quantitative phytochemical composition of the seed extract of *DE* indicate it is a rich source of carbohydrate, moisture, nitrogen, lipid, fiber and ash and contain numerous phytochemical compounds. The reported proximate and phytochemical compositions in this study compares favourably with other studies [19-21].

The non-toxic effect observed at dose of 5000mg/kg body weight following *DE* ingestion in the acute toxicity study showed the extract is safe. This assertion is made since no the dose of 5000mg/kg did not cause any alteration considering that it is the maximum allowable dose by the Organization for Economic Cooperation and Development (OECD) guideline for testing of chemicals [22]. The absence of signs of toxicity such as salivation, loss of hair, changes in animal eye color, decreased respiratory rate and motor activity as reported by Iniaghe *et al.* [23] as signs of deleterious effects of extracts, further

confirmed the ethanolic seed extract of *DE* to be safe.

Haematological parameters are important markers in the assessment of the quality and suitability of ingredients [24] and are major indices of physiological, pathological, and nutritional and metabolic status [25]. Assessment of hematological parameters can be used to determine the extent of deleterious effect of plant extracts and compounds [26] and can explain blood relating functions of chemical compounds/plant extract [27]. In the present study, exposure to CCl_4 cause a significant decrease in RBC, HCT, platelet, MPV and lymphocytes counts but significant increase in WBA and neutrophil count. These alterations from baseline levels following CCl_4 exposure is in agreement with the findings by Rahmouni *et al*., [28]. According to Essawy *et al*., [29], the decrease in RBCs count and HCT could be attributed to disturbed hematopoiesis, destruction of erythrocytes, and reduction in the rate of their formation and/or their enhanced removal from circulation. Adebayo et al. [30] reported reduced platelet count in CCl4-treated rats, leading to thrombocytopenia which tallies with the findings in this work. This reduction may be due to decreased production of thrombopoietin. Also platelets are rapidly deployed to sites of injury or infection, and potentially modulate inflammatory processes by interacting with leukocytes and by secreting cytokines, chemokines and other inflammatory mediators [31].

In the present study, the administration of the ethanolic seed extracts of *DE* returned the haematological parameters toward baseline level. This haematological parameters restorative potential support the use of *DE* in several disorders and as anti microbial, anti oxidant and anti sickle cell anaemia [13, 14]. This is expected considering that *DE* is a rich source of a wide range of bioactive compounds [12] and phytochemicals [32]. However, a study by Ufelle [33] revealed decreased Hb and Hct and increased TWBC and platelets after oral administration of graded-doses of DE seeds extract in Wistar rats. Another study by Akunne and Orhue [34] showed that extracts of *DE* stem bark and seed at high dose possess hemolytic activities and could lead to the destruction of the immune system.

CONCLUSION

From the results obtained in this study, *Dacryodes edulis* seed is rich in important nutrients and phytochemicals. The Seed of *Dacryodes edulis* when administered through the oral route appears to be practically safe with LD_{50} above 5000 mg/kg body weight. On the blood, treatment with the extract of the ethanolic seed of *Dacryodes edulis* cause improvement toward baseline in carbon tetrachloride induced heamatological alterations.

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